Graphic Era deemed to be University DEHRADUN

PROJECT AND TEAM INFORMATION

Project Title

Campus Skill Swap Network

Team Name: *Team #*	*Skill Sync* *T178*
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PHASE-II PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract (2 pts)

The Campus Skill Swap Network is a project built using C++. This project helps students learn skills from other students on their college campus Many students have abilities like programing, graphic designing and video editing but they ofen have trouble finding people who can help them learn.

Our project wants to make this easier. It allows students to create profiles. They can list the skills they can teach They can also say which skills they want to learn. The system helps them find the right people to learn from based on what they need and what they like.

A main feature is Smart Multi Way Skill Exchange. This allows three or more students to exchange skills when there is no match. The project uses C++ and Object-Oriented Programming. It also uses Data Structures to manage student information and find the matches. It also supports exchange requests and feedback. Overall the project encourages students to share knowledge learn together and develop skils through teamwork.

Updated Project Approach and Architecture (2 pts)

We have developed Campus Skill Swap Network using the C++ programming language and Object-oriented approach. Our software is designed to help students identify other students who can teach them new skills.

Students can manage their profilles, enter their skils, and indicate which skills they want to acquire. Our application scans this data and recommends students who can teach particular skills.

We have used the graph theory. It can help us impleement BFS and DFS algorithms to show connections between students and find the way to offer skill exchange. We have also applied hashing to store and find the required students quickly. Another data structure that can be used is a priority queue to store the list of students in the order of their relevance. Finally, we have decided to use SQLite as a database to store student data, skills, requests, and feedback. We have spilt the application into several modules to make it easier to manage and maintain.

Our application allows students to make direct skill exchange between two students and skill exchange between multiple students. The main purpose of our application is to help students learn new skills and share their knowledge..

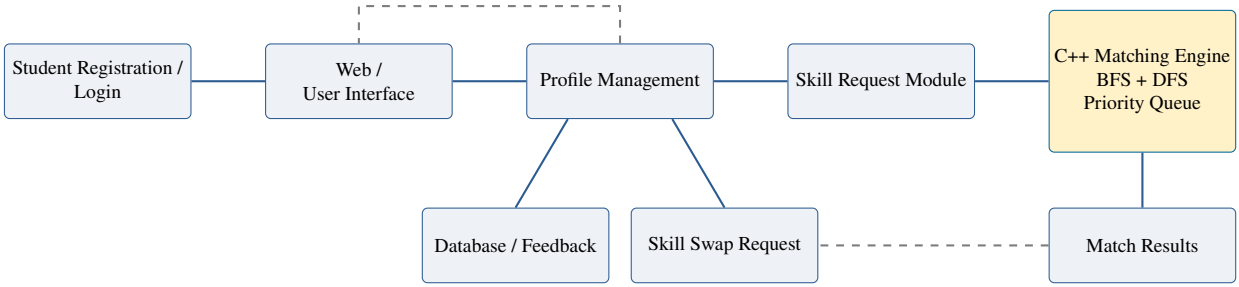


Figure 1: System architecture of the simulator

Tasks Completed (7 pts)

Task Completed	Team Member
User Registration , login , profile creation and Profile management module	Muskan Garg
Skill Management and skill search module , includeing teach or learn skills and search filters	Pranav Uniyal
Matching and Connection module	Mayank Rawat
Match Ranking and graph - based skill connections(BFS/DFS	Mayank Rawat
User interface and integration of project modules	Vaibhavi Saklani

Challenges/Roadblocks (7 pts)

1. Skill Matching:Finding students who have skills that match what others are looking for is not easy. Every student has learning goals and interests. We need to create a way that connects them effectively.

2. Multi-Way Skill Exchange:Connecting than two students in a single skill exchange is hard. One student might teach a skill to another in return for a skill. We need to make sure the logic, behind these exchanges works properly.

3. Data Structure Selection:Choosing the data structures is very important. We must understand how graphs, hash maps and priority queues function. Using them well takes time and careful planning.

4. Database Management:Storing student details and skill information correctly is difficult. We need to handle profiles, skill requests and feedback. Connecting the database to our C++ application adds another layer of complexity.

5. Match Ranking:Finding matches and ranking them in the best order is not simple. Each student has preferences and needs. We need a method to rank the matches fairly.

6. Testing the System:Testing the system with different student profiles can take a long time. Sometimes a student may not find a match at all. We must be ready to manage these cases.

Tasks Pending (7 pts)

Task Pending	Team Member (to complete the task)
Database integration for storing student profiles and skills	Muskan Garg
Saving and loading user information from the data base	Vaibhavi Saklani
Storing skill swap request , connection and feedback	Pranav Uniyal
Final testing and integration of database with all modules	Mayank Rawat

Project Outcome/Deliverables (2 pts)

The project is about creating a Campus Skill Swap Network. This network helps students who want to teach and learn skills. It uses C++ and Data Structures to find the matches and help with exchanging skills in more than one way.

The main deliverables are:

- **C++ Application:** Builds the system that handles student profiles skills that are available skills that are needed and requests for exchanges
- **Object Oriented Programming:** Uses classes and objects to keep student details and different parts of the system organized.
- **Data Structures:** Uses graphs to show how students are connected , about skills and priority queue to rank matches
- **Skill Matching Algorithm:** Finds students who're a good match based on the skills they have and the skills they need. It also helps with exchanges that involve than two students.
- **Database Management:**Uses SQLite to keep student profiles, skills, exchange requests and feedback safe and easy to manage.
- **Validâtîon:** evâluâtes how well the matching system works how students âre connected and how exchange requests âre handled

Progress Overview (2 pts)

Campus Skill Swap Network âs ân âpplication that connects students who want to learn skills with students who can teach skîls The project uses C++ and Data Structures to create the matching system based on sftudents skills that they offer and skîlls that they needd The Campus Skill Swâp Network application also allows students to exchânge skîlls in both direction

The âpplication uses Object-Oriented Programming methods to save and manage students data and specific parts of the âpplication grâphs along with algorithmss like BFS and DFS will be used to create connections.Also Hash maps and priority queues wîll be used to arrange the dâta, încluding students skills and the matching process , .Fînally SQLite will keep students dâtâ their skîls, exchange requests and feedbâck

The project focuses on making skill sharing easier within the campus. The next steps înclude developing the matching system, connenccting the database, and testing different matching sîtuations The system will be tested to check whether it can find suitable students and manage skill exchange requests properly.

Codebase Information (2 pts)

Repository Link: `https://github.com/mayankrawat5/Campus-skill-swap-`

The project is built using C++. and follows a modular approach to manage student profiles, skills, and skill exchange requests. It uses Object-Oriented Programming and Data Structures to organize information and find suitable matches between students.

Core Architectural Components:

- Object-Oriented Programming for classes nd objects to keep track of student information and skill information.
- Graph Data Structure for Showing how students are connected for exchanging skills in ways.
- Hash Maps to store and get skill information quickly.
- Graph Algorithms to look at student connections and possible ways to exchange skills.
- Priority Queue to sort and pick the skill matches.

Testing and Validation Status (2 pts)

Test Type	Status (Pass/Fail)	Notes
C++ Program Stucutre	Pass	Basic project structure and main components have been identified..
Data Sturcture Selection	Pass	Graphs, hash maps, and priority queues have been selected for the matching system..
C++ Matching Logic Design	Pass	The basic approach for matching students based on their skills has been planned..
Full and partial order matching	Pass	Verify open orders and executed quantities.
Graph Algorithms (BFS and DFS)	In progress	Working on graph traversal for multi-way skill exchange..
Database Integraton	In Progress	Working on storing and retrieving student profiles, skills, and exchange requests using SQLite.

Deliverables Progress (2 pts)

C++ project structure and oop completd: we have set up the project structure using classes and objects. This helps keep student profiles, skills and exchange requests organized and easy to manage.

Skill matching Logic completed: we have worked out how to match students based on what skills they offer and what skills they want to learn. The core idea is ready.

Data Structure completed: Priority queues for storing skill data and ranking matches. They should help make the matching faster and more efficient.

Graph Represntation Completed: the graph to show connections between students. This will help with -way skill swaps, where more than two people can exchange skills. **Graph algorithm in progress:** we r working on BFS and DFS to explore paths, between students. These will help find chains of skill exchanges.

Data base integration in pending: we r adding SQLite to save and load student profiles, skills, requests and feedback. This will make the system more real and useful.

Validation In Progress: Once the main parts are done we will test everything. I want to make sure matches work and multi-way exchanges are correct.

Final Report Pending: To be written after testing is complete.